DEPI | DATA ANALYSIS TRACK

Literature Review

Credible sources informing the project variables, methods, and interpretation

Air Quality and Life Expectancy: Exploring the Relationship Between Air Pollution and Population Health

Team Members

Ganat Saad Abdullah Hussein Ahmed | **Team Leader**
Shaimaa Maghraby Mohamed Houssien Elagouz
Habiba Maged Abdel Baqi
Dina Farag Mohamed Mohamed
Nada Ramzy Mohamed Ghnam

Prepared: 15 October 2026 | Final project target: 30 November 2026

1. Review Objective

To establish the relevance of ambient particulate matter to population health, understand how PM2.5 and PM10 indicators are defined and reported, and identify the limitations of combining city-level air-quality observations with country-level annual health data.

2. Purpose of the Review

This review examines credible sources on the health effects of particulate matter, the WHO's air-quality data collection methods, and previous research on the relationship between air pollution and life expectancy. The findings will guide the selection of project variables, data preparation, analytical methods, and interpretation of results while highlighting data coverage limitations and the difference between association and causation.

3. Reviewed Sources

Source 1: WHO Global Air Quality Guidelines (2021)

Reference: World Health Organization. (2021). WHO global air quality guidelines: particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide. Geneva: WHO.

Link: https://www.who.int/publications/i/item/9789240034228

Summary

The WHO guidelines synthesize evidence on health effects associated with major ambient air pollutants, including PM2.5 and PM10. They describe particulate pollution as a health concern and discuss associations with cardiovascular and respiratory disease and premature mortality. The document provides health-based guideline levels intended to support public-health action.

Relevance to this project

- Supports selecting PM2.5 and PM10 as the main air-quality variables.
- Provides public-health context for examining life expectancy and adult mortality.
- Shows why concentration units and careful interpretation matter.

Project note

The guidelines provide context but do not establish a statistical relationship in the team’s matched dataset. Any guideline values shown in the project must be labeled as reference levels, not as evidence of a specific health effect in an individual country-year.

Source 2: WHO Ambient Air Quality Database, 2018 Update

Reference: World Health Organization. (2018). Ambient (outdoor) air quality database: update 2018. WHO Global Health Observatory.

Link: https://www.who.int/data/gho/data/themes/air-pollution/who-air-quality-database/2018

Summary

The WHO page describes the 2018 database as a compilation of ambient air-quality measurements from human settlements across many countries. It reports annual mean PM10 and PM2.5 measurements, with coverage varying geographically and across settlements. The database documentation highlights that the available measurements do not provide uniform coverage everywhere.

Relevance to this project

- Confirms that the supplied air-quality file contains city/settlement-level observations rather than a complete national exposure measure.
- Supports aggregating city observations by country-year while retaining the number of cities contributing to each average.
- Highlights the need to check temporal coverage and whether values are measured or converted.

Project note

The supplied workbook is a specific 2018-version file, with observations from 2008 to 2017. The analysis must use the fields actually present in that workbook; it does not contain an NO2 concentration field. Only overlapping country-year records can be included in the final merge.

Source 3: Apte et al. (2018), Ambient PM2.5 Reduces Global and Regional Life Expectancy

Reference: Apte, J. S., Brauer, M., Cohen, A. J., Ezzati, M., & Pope, C. A. III. (2018). Ambient PM2.5 reduces global and regional life expectancy. Environmental Science & Technology Letters, 5(9), 546–551. https://doi.org/10.1021/acs.estlett.8b00360

Link: https://pubs.acs.org/doi/10.1021/acs.estlett.8b00360

Summary

This peer-reviewed study estimated life-expectancy reductions attributable to ambient PM2.5 for 185 countries using Global Burden of Disease data and life-table methods. The authors reported variation across regions and noted the role of background disease rates. The paper connects pollution exposure with population-level life expectancy using a more advanced modelling approach than the descriptive analysis planned for this project.

Relevance to this project

- Supports life expectancy as a population-level outcome for air-pollution research.
- Suggests examining country differences rather than relying only on a global average.
- Highlights that health conditions and population context may affect observed patterns.

Project note

The paper uses different data and modelling methods. Its numerical estimates must not be transferred to the team's dataset or presented as project findings. The current project will examine associations in a smaller, unevenly matched sample.

Source 4: Landrigan et al. (2017), The Lancet Commission on Pollution and Health

Reference: Landrigan, P. J. et al. (2017). The Lancet Commission on pollution and health. The Lancet, 391(10119), 462–512. DOI: 10.1016/S0140-6736(17)32345-0

Summary

The Commission reviewed evidence on the health and economic burden of pollution, including air, water, and soil pollution, drawing on information from the WHO and other research bodies across about 130 countries. It estimated that pollution was associated with more than 9 million deaths worldwide in 2015, with air pollution accounting for the largest share.

Relevance to this project

- Provides broader public-health context for investigating air pollution and population health.

- Supports comparing patterns across developed and developing countries, where data coverage and health burdens may differ.
- Highlights that missing observations and uneven data availability should be reported as limitations.

Project note

The Commission covers multiple forms of pollution and uses a broader evidence base than this project. Its global estimates must not be presented as findings from the team's matched dataset.

Source 5: Pope, Ezzati & Dockery (2009), Fine-Particulate Air Pollution and Life Expectancy in the United States

Reference: Pope, C. A., Ezzati, M., & Dockery, D. W. (2009). Fine-particulate air pollution and life expectancy in the United States. New England Journal of Medicine, 360(4), 376–386.

Summary

This study examined changes in PM2.5 and life expectancy across 211 counties within 51 U.S. metropolitan areas over two periods. Regression models accounted for socioeconomic, demographic, and other factors. A decrease of 10 $\mu g/m^3$ in PM2.5 was associated with an estimated 0.61-year increase in life expectancy.

Relevance to this project

- Supports examining changes in pollution alongside changes in life expectancy over comparable periods.
- Highlights the importance of considering socioeconomic and demographic factors when interpreting observed relationships.
- Provides a published example of regression-based analysis, while recognizing that the study population and methods differ from this project.

Project note

The reported estimate comes from U.S. counties and a specific study design. It is a contextual reference only and must not be treated as an expected result or directly transferred to the team's country-level dataset.

Source 6: Chen et al. (2013), Evidence on the Impact of Sustained Exposure to Air Pollution on Life Expectancy from China's Huai River Policy

Reference: Chen, Y., Ebenstein, A., Greenstone, M., & Li, H. (2013). Evidence on the impact of sustained exposure to air pollution on life expectancy from China's Huai River policy. Proceedings of the National Academy of Sciences, 110(32), 12936–12941. DOI: 10.1073/pnas.1300018110

Summary

The study used a quasi-experimental design based on a policy that created differences in coal-based heating access north and south of China's Huai River. It compared total suspended particulate (TSP) pollution and life expectancy near the policy boundary. The researchers reported higher TSP concentrations and lower life expectancy north of the river.

Relevance to this project

- Shows why establishing causality requires stronger designs than a simple comparison or correlation.
- Supports careful wording such as "is associated with" when interpreting this project's observational data.

- Highlights the need to distinguish pollutants and units, since the study measured TSP while this project focuses on PM2.5 and PM10.

Project note

The study's quasi-experimental design and pollutant measure differ from the project's methods and variables. Its numerical estimates should not be compared directly with the project's results.

Source 7: Cohen et al. (2017), Estimates and 25-Year Trends of the Global Burden of Disease Attributable to Ambient Air Pollution

Reference: Cohen, A. J., Brauer, M. et al. (2017). Estimates and 25-year trends of the global burden of disease attributable to ambient air pollution: an analysis of data from the Global Burden of Diseases Study 2015. The Lancet, 389(10082), 1907–1918.

Summary

Using data from the Global Burden of Diseases Study 2015, this research estimated deaths and disability-adjusted life years attributable to ambient air pollution across countries and over time. It reported that PM2.5 was a major global risk factor in 2015, with substantial regional differences in the health burden.

Relevance to this project

- Supports examining pollution trends over time as well as differences between countries and regions.
- Reinforces the relevance of PM2.5 as a key indicator in the project's analysis.
- Highlights that population characteristics and underlying disease patterns can influence health outcomes alongside pollution.

Project note

The paper estimates disease burden using a different dataset and methodology. The project can use its approach as background for choosing analyses, but must not treat its global estimates as results from the supplied workbooks.

4. References

- Apte, J. S., Brauer, M., Cohen, A. J., Ezzati, M., & Pope, C. A. III. (2018). Ambient PM2.5 reduces global and regional life expectancy. Environmental Science & Technology Letters, 5(9), 546–551. https://doi.org/10.1021/acs.estlett.8b00360
- World Health Organization. (2018). Ambient (outdoor) air quality database: update 2018. https://www.who.int/data/gho/data/themes/air-pollution/who-air-quality-database/2018
- World Health Organization. (2021). WHO global air quality guidelines: particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide. https://www.who.int/publications/i/item/9789240034228
- Kaggle. Life Expectancy (WHO) dataset. https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who
- Kaggle. Ambient Air Quality Database. https://www.kaggle.com/datasets/fkosmowski/ambient-air-quality-database
- Landrigan, P. J. et al. (2017). The Lancet Commission on pollution and health. The Lancet, 391(10119), 462–512. https://doi.org/10.1016/S0140-6736(17)32345-0

- Pope, C. A., Ezzati, M., & Dockery, D. W. (2009). Fine-particulate air pollution and life expectancy in the United States. New England Journal of Medicine, 360(4), 376–386.
- Chen, Y., Ebenstein, A., Greenstone, M., & Li, H. (2013). Evidence on the impact of sustained exposure to air pollution on life expectancy from China's Huai River policy. Proceedings of the National Academy of Sciences, 110(32), 12936–12941. https://doi.org/10.1073/pnas.1300018110
- Cohen, A. J., Brauer, M. et al. (2017). Estimates and 25-year trends of the global burden of disease attributable to ambient air pollution: an analysis of data from the Global Burden of Diseases Study 2015. The Lancet, 389(10082), 1907–1918.